Comparison Multi Transfer Learning Models for Deep Fake Image Recognizer

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***Abstract*— The advancement of technologies nowadays cause image digitally to become essential as an official document many available applications used for image editing. Those applications have become a threat to detect image authenticity when dealing with an abundance of digital image evidence in cyber court. Hence, many researchers realize the importance of image authentication fields as deep fake is a powerful weapon for spreading misinformation on the digital platform. Deep learning has known to obtain relevant attributes automatically in placing handcrafted features in a deep network against other single-layer networks. The objective of the research is to compare two models from Convolutional Neural Network (CNN), which are VGG19 dan ResNet50 in deep learning with transfer learning for image fake detector. A total of 1500 random images consisting of 450 forgery images from the IEEE Image Forensics Challenge in 2013 were tested on the splicing technique. From this research, we used two transfer learning techniques to identify tamper images from image splicing. Based on model VGG16 and ResNet50 transfer learning, the accuracy achieved about 94.65% and 95.08%, respectively.**

Keywords—deep learning, neural network, deep fake a transfer learning

I. INTRODUCTION

Digital 2019 Reports reported that more than 4.39 billion internet users in the year 2019, which shows an increase of 366 million users since January 2018. Another intriguing statistic mentioned by the Digital 2019 Reports is that there are 3.4 billion users of social media such as Facebook and Instagram (Hootsuite & We Are Social, 2019). With the increase of social media users, there is also an increase in image editing applications such as Lightroom and Adobe Photoshop because most social media requires an image to be uploaded. It has proven that social media tends to cause body image concerns, which lead to issues like eating disorders and photoshopping images (Bell, 2016). More editing applications emerge, and the amount of altered digital images increases on the internet, which requires forensic image analysis. Plenty of methods and techniques that can determine the authenticity of a digital image are being discovered and researched (Mizher et. al, 2017; 2019). However, the usage of digital images is increasing at an alarming rate and currently are also being used in official documents for most occupations. Digital images are also being used as evidentiary support in legal cases; thus, more accurate forensic analysis methods are needed (Ghazali et. Al., 2012; Abdullah et. al, 2017).

Deep learning has been known to perform well in handling images and achieving high precision in object detection, object counting, and image classification. It utilizes a very deep neural network model such as Residual Neural Network (ResNet), VGGNet, Inspection model, and others. There is a well-known deep learning technique called transfer learning, which can be used to develop a neural network model by reusing previously learned knowledge from the existing pre-trained model (Pan & Yang, 2009). This is possible because a deep neural network is known to be able to learn transferable features. Thus, in this paper, two neural network models, namely VGG16 and ResNet50 are implemented using transfer learning, and the model will then be used to detect edited images. Detection of fake images consists of proving whether the images are edited and highlighting the edited region in the images.

II. LITERATURE REVIEW

Digital image forensic (DIF) is a discipline that conducts image processing and analysis on digital images through tools or techniques to identify if there are any elements of alterations (Redi et al., 2011; Mizher et al., 2017;2019,). DIF also consists of identifying the timestamp and location of the device that produces the edited image. Obtaining that information is difficult because image tampering tends only to leave a slight trace, which makes tracking difficult. Tampering an image occurs when the semantic content of an image has changed due to an addition or removal parts of an image (Redi et al., 2011). Detection of those tampered

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regions follows the general pipeline, namely feature extraction, feature matching, and post-processing (Cozzolino et al., 2015). In this section, two main topics are the techniques of image alteration and the performance of utilizing the deep network in performing DIF.

Digital image processing (DIP) is a process of image alteration through programming or computer software to enhance, edit, analyze, and improve the quality of an image. DIF can identify those changes and assure that the image content is not altered. In general, there are two categories of digital forgeries, namely an active approach and a passive approach (Parashar et al., 2015). The active approach is not the primary concern as it is not able to identify the authenticity of a digital image. There are three main methods for the passive approach, which are copy-move, image splicing, and image retouching.

A. Deep Learning Application in Forgery Detection

Based on all the forgery detection techniques stated, those techniques utilize statistical data and statistical models. However, in this paper, a deep neural network is being used to conduct the detection. Deep learning has been known to have an impressive performance in object detection and obtained a very high accuracy. Currently, there are several efforts to integrate deep learning and digital image forensics (DIF). Yuan Rao proposed a work that uses a deep learning approach to conduct splicing and copy-move detection (Rao & Ni, 2016). The proposed convolutional neural network (CNN) contains 8 convolutional layers, 2 pooling layers, and a fully connected layer with a softmax classifier that takes RGB color images as an input. The first layer contains 30 basic high-pass filters used in the spatial rich model (SRM) is applied to suppress the effect of complex image content. Compared to other CNNs used in other works, this neural network only exploits one fully connected layer due to the lack of the number of the training set. The modified CNN demonstrates outstanding performance in detecting forgeries on digital images.

Moreover, Weize Quan et al. proposed another work that exploits CNN to obtain a high accuracy model for colorized image detection (Quan et al., 2019). They have decided to develop an “end-to-end” framework based on CNN to distinguish between natural and colorized images. This framework is also known as a unified framework where both recurrent neural networks (RNNs) and CNN's are combined (Wang J. et al., 2016). Thus, this framework learns the embedded joint image-label and model the label co-occurrence dependency in a joint label embedding space. The base network of CNN has improved by adding a new branch to attain better performance in the generalization capability. Another step in improving forensic performance is by including an additional auto-constructed negative sample into the original training dataset. Further and enhanced training of the network is then conducted for better generalization performance. With the presence of several precaution steps in improving generalization, the output model shows stable performance in handling blind detection scenarios; however, this approach reduced the classification accuracy slightly.

Another approach was conducted by Jason Bunk et al., which utilizes two methods to perform forgery detection (Bunk J. et al., 2017). The first step uses an end-to-end framework based on Radeon transform and deep learning, which localized the manipulated region. The following method applies resampling features from probability-maps and combining it with Long short-term memory (LSTM) model to classify altered patches. The resampling method requires interpolation, which presents periodic correlations among pixels (Mohammed et al., 2018). A linear or a cubic model can detect this correlation. But, in this work, a linear approach is being used. This work intended to identify the accuracy of both methods and comparing the results. It is safe to say that both methods perform well in detecting manipulated regions by utilizing resampling features.

B. Transfer Learning

In this work, the well-known deep learning technique, namely transfer learning, is used to develop the detection model. In simpler words, transfer learning allows the current model to learn from the previous model by transferring knowledge. Generally, there are two types of transfer learning: heterogeneous transfer learning and homogenous transfer learning (Weiss et al., 2016). Heterogeneous transfer learning is when the source project or model does not contain the same features as the target project, while homogenous transfer learning has the same values for both source and target models. However, another term is present with regards to transfer learning, which is called negative transfer. Negative transfer happens when the knowledge transferred from the source domain produces a negative impact on the target source due to the level of relativity between both sources. Thus, to obtain an accurate detection model, the transfer learning process needs to project a positive transfer from the source to the target model. This behavior can ensure that both the source and the target model features are related to one another. Deep networks such as AlexNet, ResNet, VGGNet, Inception and Xception have been regularly utilized by applying transfer learning.

III. METHODOLOGY

For the coding developed in this study, CNN's in-depth learning method proposed by Rao Yuan and Ni Jiangqun (2016) has been adopted into our CNN design proposal. Fig. 1 below shows a CNN design containing eight convolutional layers, two pooling layers, and a fully interconnected layer together with two-way softmax classification. The number of inputs is $128 \times 128 \times 3$. However, the above design with $128 \times 128 \times 3$ input volume is too large for this study. Accordingly, we changed the CNN's design by subtracting half of the above.

Fig. 1 below shows a modified CNN design with an input of half of the original, $64 \times 64 \times 3$. This design means the input for the CNN model in this study is 64 height×64 width $\times$ 3 channels. The 3 channels represent red, green, and blue

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(RGB). We selected this design by following the data set capacity used.

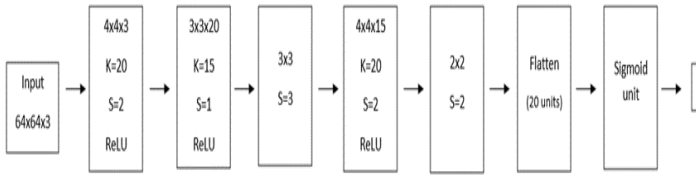


Fig. 1. Suggested CNN Model for Deep Fake detection.

A. Idea of the transfer learning method

Conventional machine learning and deep learning approach are often designed in separate modules. The transfer learning technique was then introduced to overcome the isolated learning paradigm. This technique is defined as applying the knowledge produced from past trained models (pre-trained) to solve a different yet related problem. The idea behind the transfer learning approach is shown in Fig. 2 below. In the transfer learning environment (Pan & Yang, 2009), the source domain will be represented as D_S and the source task is indicated as T_S. While the target domain is indicated as D_T and T_T as the target task. The main objective of transfer learning is to enhance the acquisition of knowledge of the target predictive function, $f_T(\cdot)$ which can also be represented as a probability distribution, $P_T(X)$, where X is the feature space. This can be done by utilizing the knowledge extracted from D_S and T_S.

Estimation of H-divergence (Jin et al., 2020) between source and target domain is defined in (1) which will aid in calculating the divergence of unlabeled data. H represents the hypothesis class on feature space X and the set for the characteristic function is notated as $I(h)$,where $h \epsilon H$.

$$d_H(D_S, D_T) = 2 \sup_{h \in H} |P_S[I(h)] - P_T[I(h)]| \quad (1)$$

Moreover, the relationship between errors of target domain, e_t and source domain, e_s can be defined based on equation (2) below. Let m' be the unlabeled sample sizes extracted from each domain.

$$e_t(h) \leq e_s(h) + \frac{1}{2}\hat{d}_{H\Delta H}(u_s, u_t) + 4\sqrt{2d\log(2m') + \frac{\log\frac{2}{\delta}}{m'}} + \lambda \quad (2)$$

where u_s and u_t are unlabeled samples obtained from source domain and target domain. In order to find the difference in errors of each domain, equation (3) is represented as follows,

$$|e_t - e_s| \approx \frac{1}{2} d_{H\Delta H}(D_S, D_T) \quad (3)$$

$d_{H\Delta H}$ is the calculated symmetric difference hypothesis space $H\Delta H$ that is beneficial for reasoning purposes. It is evident that transfer learning is applicable in various domains and is mathematically effective based on the equations stated. This proposed study aims to perform forgery detection thus, the source task is object detection, and the target task is tampered image detection.

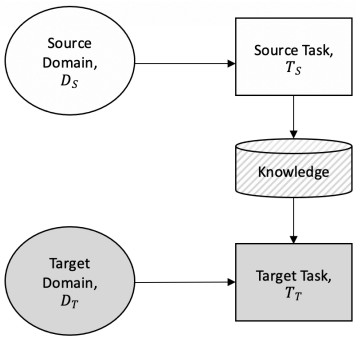


Fig. 2. Learning Process of Deep Transfer Learning

In this study, the deep learning methods have taken VGG16 and ResNet50 models as the study method (Sarkar D.J, 2018). The transfer learning methods used in this study are as follows:

- Uses the bottleneck features from the pre-training series combine with the shallow series.
- Performs fine-tuning on the last layer combine with the superficial network (the fine-tuning model)

The usage of the final convolution layer results in VGG16 and ResNet50 is also known as the feature input to the additional layer. The additional layer is known as the CNN superficial layer. This shallow layer input has only one channel—the design of the CNN shallow layer, as illustrated in Fig. 1 in the last three layers.

The last layer selection in this study was to avoid overfitting as the whole network on the VGG16, and ResNet 50 designs had very high entropy capacity. In this study, the modifications to the last layer use very low learning rates so as not to damage the other learning features.

The amount of data used is not the CASIA data set used in previous studies. For the convolution layer, the ReLu activation function was subsequently employed. The activation function equated to the amount of weight that is placed on the input to determine which input can be processed. The use of ReLu as a function of activation in this study is because ReLu provides a linear computational function, so without complicated calculations, the learning model takes lesser time for data testing or training. Layers are known as pooling, and max-pooling layers are used as settings. This layer functions to lower the input size by modifying the new input scale.

B. Set of Data

During the first Image Forensic Challenge in 2013, the IEEE committee provided several digital image data sets with the following types and counterfeiting techniques: Content-Aware Fill and PatchMatch (for copy/pasting), Content-Aware Healing (for copy/pasting and splicing), Clone-Stamp (for copy/pasting), Seam carving (image retargeting), Inpainting (image reconstruction of damaged parts — a special case of copy/pasting) and Alpha Matting (for splicing). Alpha Matting digital image data set contains about 1050

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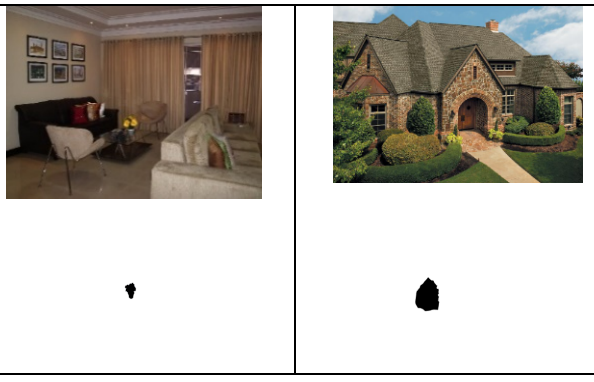
pieces of a digital image of the original and 450 pieces of the fake digital image along with a black and white-collar. Below are the samples of non-fake and fake images with its ground truth. Table 1 below shows the sample of images used for this paper which contains more original images than tampered images. The upper table contains non-fake images. Meanwhile, the bottom table shows the selection of fake images with the tampered region's splice in white and black colour.

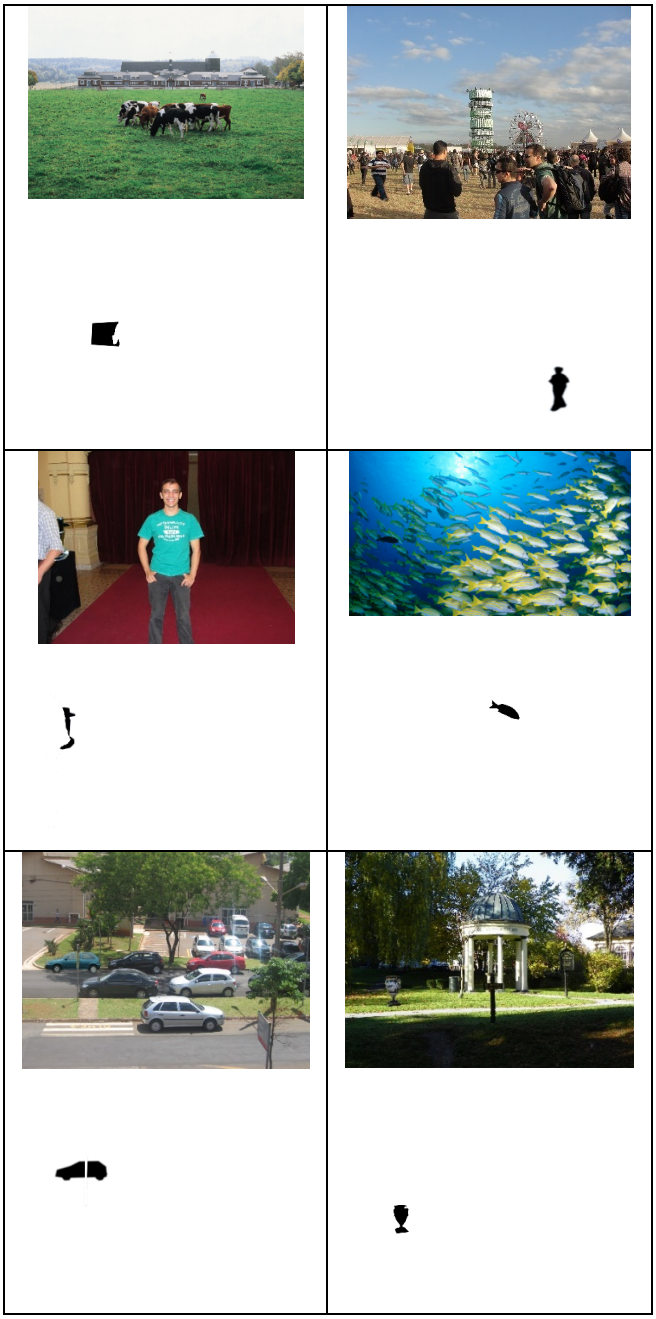
TABLE I. EXAMPLE OF (TOP) NON-FAKE AND (BOTTOM) FAKE IMAGE DATA SAMPLES

Original (Non-fake)



Fake Images





IV. EXPERIMENTS & RESULTS

The results of the study for deep learning are shown in Table 2 below. Several models have been used to compare accuracy results. It summarizes the results of the study based on the trained data set and validation data set. For the traditional CNN models, the accuracy was 77.13%, with a loss rate of 0.4678 of training data and accuracy of 75.68% and a relatively high loss rate of 0.5121.

TABLE II. RESULTS OF IN-DEPTH LEARNING AND TRANSFER LEARNING ON THE DATASET

Model	Training Dataset		Validation Dataset	
	Acc (%)	Loss	Acc (%)	Loss

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CNN Model	77.13%	0.4678	75.68%	0.5121
Transfer Learning				
Bottleneck model VGG16	83.18%	0.3230	84.26%	0.3833
Bottleneck model ResNet50	94.24%	0.1297	91.66%	0.2483
The fine-tuning model of the last layer of VGG16	82.84%	0.3282	85.25%	0.3398
The fine-tuning model of the last layer of ResNet50	93.24%	0.1351	93.08%	0.2228
Using Test Data Sets				
The fine-tuning model of the last layer of VGG16	94.65 %	0.3122		
The fine-tuning model of the last layer of ResNet50	95.08%	0.1938		

For the bottleneck model feature, the study carried out using VGG16 and ResNet50 design. Accuracy results for VGG16 were 83.18% for training data and increased by 1.08% for validation data. The loss rate for this model was 0.3230 for training data and 0.3833 for validation data. ResNet50 accuracy result for training data was 94.24%, but for validation, data accuracy decreased to 91.66%. The loss rate was 0.1297 for training data and 0.2483 for validation data. Based on those results, both models obtained accuracy of more than 80% and loss rate of lower than 0.5. Thus, a conclusion can be made that both models are relatively successful in detecting tampered images. The fine-tuning model of the last layer also uses the VGG16 and ResNet50 designs. The results obtained for the VGG16 design are 82.84% accuracy for training data and 85.25% for validation data, while loss rates for training data are 0.382 and 0.3398 for validation data. ResNet50's design study gained 93.24% accuracy for training data and 93.08% accuracy for validation data. The loss rate for training data was 0.1351 for training data and 0.2228 for validation data.

Due to time constraints, studies using test data were run on finite tuning models in the last layer using the VGG16 and ResNet50 designs. The results showed that the ResNet design gave a higher accuracy of 95.09% than the VGG16 with 94.65% accuracy and the ResNet50 loss rate was 0.1938 and the VGG19 was 0.3122. The epoch iteration set in this study was 35 because it shortened the coding process. If the number of epochs is too small, the learning process in the model may work improperly. Therefore the 35-epoch iteration in this study is sufficient for the learning process in VGG16 and ResNet50 models, and it does not take too long.

Fig. 3 (a) summarized in the model of the VGG16 bottleneck features, the accuracy results show an increase in each epoch, but at epoch 21, the resultant decomposition remains to the last epoch. This situation is very similar to the training and validation data sets. Meanwhile, the loss rate begins with low volatility in the last epoch, and the loss rate begins to decline until the last epoch.

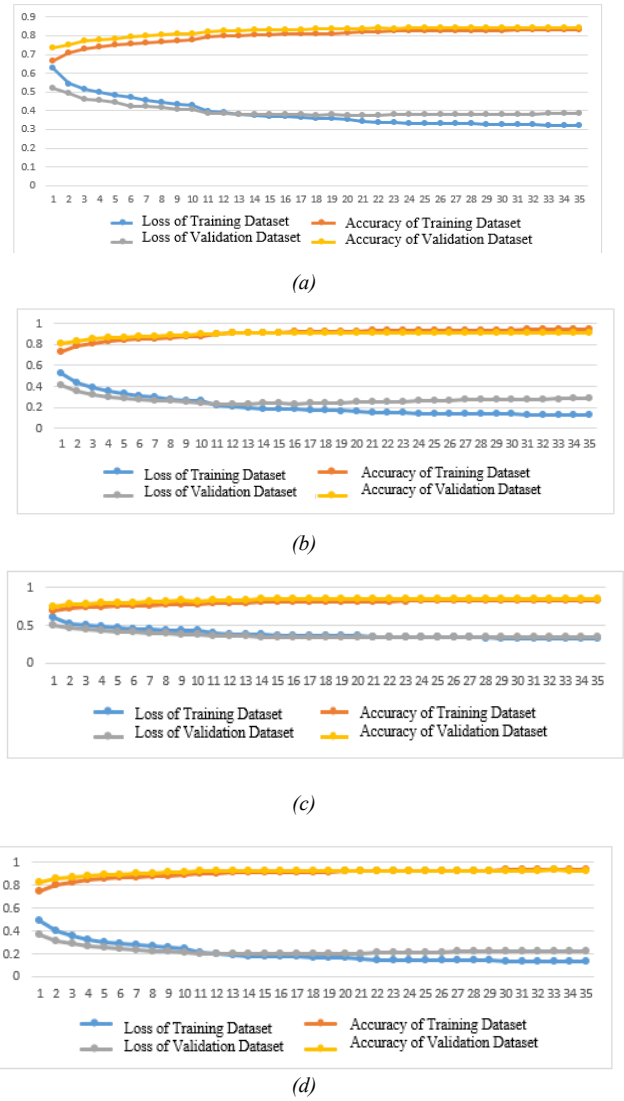


Fig. 3. The epoch iteration of Bottleneck Models for (a) VGG16 and (b) ResNet50, and Fine-Tuning model of (c) VGG16 and (d) ResNet50 subsequently.

The same situation applies to the ResNet50 cube feature model, as shown in Fig. 3(b). The results of accuracy on training and validation data sets do not have many differences from the beginning until the end of the epoch.

The epoch iteration of the fine-tuning model in the last layer of VGG16 is horizontal and has not many differences based on Fig. 3(c). This situation is very similar to the results of accuracy and loss rates for training sets and validation. For the fine-tuning model of the last layer in ResNet50 (Fig. 3(d)), the resultant accuracy was only slightly different from 0.16 for the training data set. Its accuracy results are increasing from the first epoch to the last epoch. The rate of loss also decreasing from the first epoch to the last epoch.

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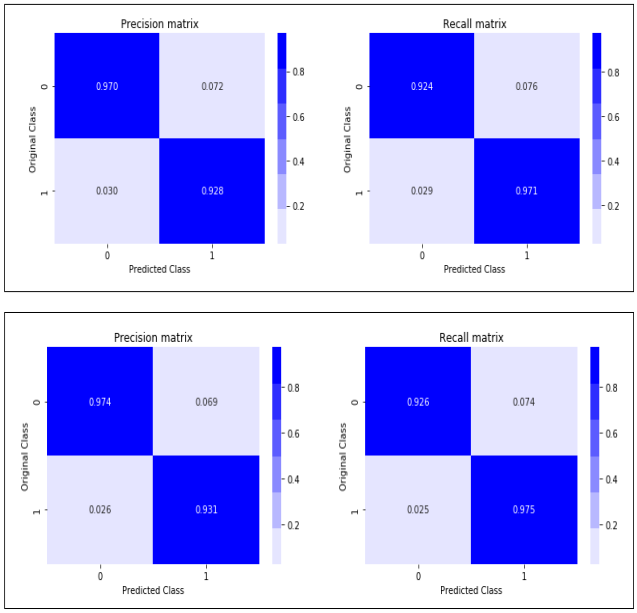


Fig. 4. Binary confusion matrix for fine-tuning models in the final layer of (top) VGG16 and (bottom) ResNet50 using the test data sets respectively.

As stated before, the number of original images is more than tempered images on this database. Therefore, Binary Confusion Matrix is used to calculate true positive (TP), true negative (TN) and loss rate in this paper. The results of the accuracy and loss rates with the performance matrix are shown in Fig. 4 for (a) VGG16 and (b) ResNet50. We can conclude that this study gave a satisfactory result because the Performance Matrix of the accuracy achieves greater than 88.96%, and the loss rate should be less than 0.4968.

V. CONCLUSION

In conclusion, the ResNet50 design produces higher accuracy compared to VGG16. This trend appears because of the deeper capabilities of the ResNet50 design network. At the same time, we are testing an unlabelled data set. Incoming Resnet can produce higher accuracy than VGG16 with a lower loss rate than VGG16. The main reason for choosing transfer learning in this study is that transfer learning can save time, cost, and it is a stable model of deep learning. The use of deep learning alone is impossible without adequate experience.

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